

# Samiul Hossen Sarkar Santo

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**Aspiring MLOps engineer with a strong foundation in software development and hands-on experience building end-to-end systems. Interested in scalable ML pipelines, Model deployment, and real-world AI applications.**

## EDUCATION

**Bangladesh University of Professionals | Mirpur Cantonment, Dhaka**

**July 2022- Present**

*B.Sc. in Information and Communication Engineering*

**CGPA:** 3.67/4.00 (Till 7<sup>th</sup> Semester)

**Relevant Courses:** Machine Learning, Artificial Intelligence, Analysis and Design of Algorithm, Software and Requirement Engineering, Object Oriented Design and Programming, Database Management System

**Extracurricular Activities:** Captain of the ICT Volleyball Team, Anchoring, Poetry recitation, Table Tennis.

**Aspire Institute | Harvard Business School**

**May 2025- August 2025**

*2025 Aspire Leaders Program*

Cohort 3

## TECHNICAL SKILLS

**Languages:** Java, Python, C, JavaScript, SQL, Dart

**ML & Data:** Pandas, NumPy, Scikit-learn, MediaPipe

**Technologies:** Git, Flutter, Firebase, Node.js, Servlet

**Tools:** Google Colab, Notion, Trello, Canva

## EXPERIENCE

**IEEE BUP Student Branch WIE Affinity**

**May 2024 - Present**

*General Secretary (October 2025 - Present)*

*Webmaster (May 2024 - October 2025)*

- Organized 15+ workshops, seminars, and national festivals (e.g., IEEE BranchFest 2025, Google/Microsoft Career events).
- Conducted a seminar, <Beyond HelloWorld!> 2.0, as the Keynote Speaker.
- Hosted multiple seminars and events (e.g., Google and Microsoft Career events, IEEE BranchFest 2025).
- Managed cross-functional collaboration by coordinating support tasks for 3 different teams and co-leading design and content teams.
- Developed strong communication and project management skills in a fast-paced, collaborative environment.

## PROJECTS

**Investigating Dialect Bias in Bangla NMT | Python, NLP, WER/CER Metrics**

**October 2025 - Present**

- Undergrad Thesis (Ongoing)
- Investigating translation bias in pretrained Bangla neural machine translation (NMT) systems by evaluating performance across regional dialects including Sylheti, Chittagong, Barishal, and Noakhali.
- Curating and harmonizing a balanced parallel dialect dataset (~4,500 sentence sets) by integrating multiple research corpora with strict filtering, alignment, and manual validation.
- Evaluating model performance using Word Error Rate (WER) and Character Error Rate (CER) while applying statistical significance testing to quantify dialect-wise translation degradation and bias severity.

**Sign Language Detection | MediaPipe Hands, KNN, IndexedDB**

**December 2025**

- Engineered a browser-based American Sign Language (ASL) translator integrating MediaPipe Hands and a lightweight KNN classifier to learn and detect gestures in real-time.
- Built local dataset management via IndexedDB to capture custom samples, preview recorded poses, and load a bundled starter dataset (asl\_dataset.json) for immediate use.
- Implemented accessibility and export features to seamlessly convert recognized text into automated speech or downloadable PDF documents.

**Thesis Data Assistant | HTML, JavaScript, Tailwind CSS**

**January 2026**

- Streamlined dataset preparation workflows to support reproducible ML experimentation.
- Created a privacy-first, client-side web utility to assist academic researchers with local data parsing, formatting, and file conversion (CSV, Excel, JSON).

- Incorporated OCR capabilities using Tesseract.js to accurately extract English and Bengali text directly from uploaded images without requiring server-side processing.

#### [BUP Bus Tracker](#) | Flutter, Firebase, Google Maps API

**December 2024**

- Developed a real-time transit tracking application to facilitate daily commutes for university students by visualizing live bus locations.
- Integrated Google Maps API and Firebase to provide accurate spatial data mapping and seamless, real-time database synchronization.
- Serving 500+ active users currently.
- Integrated with BUP Bus Tracker- Driver App to evade physical GPS tracker dependency.

#### [BUP Bus Tracker - Driver App](#) | Flutter, Google Maps API

**December 2024**

- Built a companion mobile application enabling drivers to broadcast live GPS coordinates to the primary student tracking system.
- Optimized location updates to ensure low latency tracking performance while minimizing mobile device battery consumption.

#### [Leave It Here](#) | Flutter, Local Storage (File System)

**March 2026 - Present**

- Developed a privacy-first journaling app with guided reflection, breakdown tracking, and mood-aware wins extraction in a local-first architecture.
- Implemented voice journaling with in-app recording and playback, including voice-only entry support and seamless integration into the entry workflow.
- Designed intuitive UI/UX focused on low-friction writing experience for emotionally heavy days.
- Built app security features with PIN/biometric lock, lock timeout controls, and protected access flows for sensitive personal entries.
- Engineered offline backup and restore for full cross-device migration (entries, settings, and voice notes) via export/import without requiring accounts or cloud services.

#### [Event Koi?!](#) | Node.js, Express.js, MongoDB

**May 2025**

- Designed a centralized web platform for discovering, managing, and registering for nationwide events and competitions.
- Built role-based dashboards for participants, organizers, and administrators with secure authentication.
- Developed RESTful APIs using Express.js to handle event workflows and user interactions.
- Structured and optimized MongoDB schemas to support scalable data management and efficient query performance.

#### [Chopsticks](#) | Java

**April 2024**

- Programmed a console-based adaptation of the classic hand game by strictly leveraging Object-Oriented Programming (OOP) principles.
- Implemented game state logic, turn-based mechanics, and dynamic rule validation to ensure an interactive and bug-free user experience.

#### [Vocab Victory](#) | Java

**June 2024**

- Constructed a console-based educational game designed to test and dynamically expand user vocabulary through interactive text challenges.
- Applied OOP methodologies to structure game components modularly, promoting code reusability and streamlined testing procedures.

#### [Crafty Cat](#) | Java, JSP, Servlets, MySQL

**September 2024**

- Developed a fully functional e-commerce web application featuring user registration, product catalog browsing, and dynamic shopping cart management.
- Managed relational data schemas using MySQL and handled robust server-side logic with Java Servlets to process transactions securely.